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Accelerating Molecular Generation through Representation Alignment

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Abstract

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Chapter 1

Introduction

The evolution of generative machine learning in scientific domains is defined by the tension between data scale, data quality, and structural priors. Our report examines a recent addition to this story in *representation alignment*. We study this technique in two regimes for de novo molecular generation — small molecules and protein backbones — and characterise its behaviour. In this introduction, we trace the narrative arc of research trends in molecular generation, and highlight the motivation for our work: the emerging need for data-efficient modelling in a world of prior-light models.

The dominant ideas in generative modelling are *denoising diffusion* and its offshoot *flow matching* [1, 2], which found breakthrough success in the realm of image generation. This paradigm has since been adapted for de novo design challenges across scientific fields, such as generating protein backbones [17, 18, 13], inorganic materials [20], and drug-like small molecules [14]. Training these models is typically expensive in both data and compute, with state-of-the-art image generators routinely training on billions of examples [27]. However, the corpus of available high-quality molecular data sits orders of magnitude below this. Experimental determination is slow and expensive, which means datasets are correspondingly small. For example, protein structure generators have largely been trained on at most half a million structures [13], which may explain why they have not yet matched the maturity of their vision-domain counterparts.

Historically, researchers compensated for the small scale of high-quality molecular data by encoding structural priors directly into model architectures [33]. SE(3)-equivariant architectures, for example, enforce the rotational and translational symmetries expected of molecules by construction [35]. But hard structural priors come with a computational cost, and the operations required to encode geometry are a bottleneck to scaling models.

A recent wave of research has moved entirely the other way: drop hard priors, use bare-bones non-equivariant transformer stacks, and rely on data scale and augmentation to teach the model what equivariance buys. This shift fits a broader trend in machine learning on leveraging representation instead of model architecture to inject soft inductive

biases for structured data. Vision Transformers [36] showed that the spatial-locality and translation-equivariance priors of convolutional neural nets (CNNs) can be learned rather than imposed, given sufficient scale. Graph Transformers like TokenGT [37] have made the same case for graphs. AlphaFold 3 [39] stripped many of the SE(3)-equivariant biases of AlphaFold 2 [38], replacing the structure module with a diffusion model operating on raw coordinates. Brehmer et al. [40] make the empirical scaling case directly: more data through a simpler model beats less data through a richer one. In our work, we examine Tabasco [14] and Proteina [13], which are flow-based generators built on vanilla transformers for small molecules and protein backbones respectively.

Given the difficulty of procuring data in scientific domains, data scale must necessarily come from non-experimental sources, which may be of lower quality. For example, recent efforts in protein generation have leaned on synthetic structures from folding-model predictions [13, 29] to expand training datasets into the tens of millions. However, even these enhanced corpora are arguably insufficient. Metagenomic databases catalogue over two billion natural protein sequences [31], and the theoretical sequence space at a median protein length [32] is hundreds of orders of magnitude greater still. Even at the frontier of the field’s data scale, we sample but a sliver of the distribution we wish to model.

In summary, the field has traded hard structural priors for architectural simplicity, and curated data for synthetic scale. However, this trade-off dilutes the structural signal available to a model. Furthermore, even our best attempts at creating large datasets may not be large enough. These twin challenges amplify the need for techniques that make training still more efficient without sacrificing quality.

But why is training diffusion-transformer architectures expensive in the first place? One answer comes from Yu et al. [5], who argue that these models suffer from a *representation bottleneck*. During training, a model is implicitly asked to learn two distinct tasks simultaneously: (1) representation: extracting semantically meaningful encodings from complex data; and (2) generation: constructing data from these encodings. The standard denoising objective alone may not provide sufficient signal for representation learning, leading to slow convergence and the large training budgets these models require in practice. In the context of molecular generation, this representation burden is compounded by the twin challenges outlined above. Models are now forced to encode complex structural geometry from data that is vastly smaller than the true underlying distribution, all while operating without the aid of architectural priors.

Yu et al. propose Representation Alignment (REPA) as a remedy. The technique adds a regularisation term that aligns a generator’s intermediate hidden states with the embeddings of a frozen pretrained encoder. On image diffusion, they report roughly a $17.5\times$ training speed-up against a strong baseline. REPA has since been extended beyond images to video [9], audio [10], and recently to molecular generation [7, 8]. However, whether REPA’s gains transfer to molecular settings, and why, remain uncharacterised.

In this light, the core question we investigate in this report is: *can prior-light molecular generation models benefit from representation alignment with pre-trained models, in training efficiency and generation quality?* We aim to characterise when and why this works, including failure modes and limitations.

Our report documents two lines of enquiry.

- (i) *Does REPA work in this domain?*: We conduct empirical studies on training flow-based molecular generator architectures with REPA in two regimes: Tabasco [14] for small molecules, and Proteina [13] for protein backbones. We find that REPA does not measurably improve Tabasco, but it improves training efficiency and advances the generation quality envelope for Proteina.
- (ii) *When does REPA work?*: Following Singh et al. [6] in the realm of image generation, we develop a profiling framework to characterise the suitability of an encoder as a representation target. We measure three properties of an encoder: (1) an upper bound on the structural information it contains (linear-probe recoverability against domain structural labels); (2) a lower bound on what a projector can transfer from that information (projector-based recoverability); (3) the geometric conditioning of its embeddings for the alignment objective (effective rank).

We do not propose a new architecture, alignment loss, or evaluation metric, and our two studies are not a controlled experiment over the conditions that govern REPA’s effectiveness. Instead, they are best read as paired case studies in regimes where those conditions differ.

The remainder of this report is organised as follows. Chapter 2 sets out the background needed to read the rest: flow matching, REPA, and the parameterisations we use for small molecules and proteins. Chapter 3 places the work in context, covering the scaling-over-priors trend in structural biology and the recent literature on representation alignment. Chapter 4 discusses what it means for a generative model to be “good” in our domain, and the tensions that question hides. Chapter 5 develops the encoder-profiling framework and uses it to predict where REPA may and may not help. Chapters 6 and 7 present the two paired studies, small molecules first and protein backbones second. Finally, we conclude in Chapter 8 with a discussion of whether what we learn generalises beyond molecular generation.

Chapter 2

Background

In this chapter, we introduce the machinery the rest of our report depends on. We outline flow matching (§2.1), the generative paradigm underlying the models we study, and representation alignment (§2.2), the regulariser that is our object of study. We then provide some motivating context on small molecules and protein backbones (§2.3), and discuss how they are typically parameterised for generative machine learning workflows. Finally, we outline Tabasco and Proteina (§2.4), the two flow-matching backbones we extend in our experiments. The treatment is cursory by design, intended only to guide the reader through this report, and we provide citations for omitted details.

2.1 Flow matching

At first glance, the term *generative modelling* implies the problem of *creating* new artefacts from scratch. However, it is perhaps more accurately described as the problem of *sampling* new artefacts from a complex data distribution that can only be accessed through a finite training set. A common strategy here is to fix a simple distribution we can sample from cheaply — typically standard Gaussian noise $p_0 = \mathcal{N}(0, I)$ — and learn a mapping from it to the desired data distribution p_1 .

Flow matching realises this mapping as a continuous-time transport. The model designer defines a *probability path* between a noise sample and a data sample — a continuous interpolation between them — and the model learns to carry the noise sample along this path. Concretely, it parameterises a time-dependent *velocity field* $v_\theta(x, t)$ that gives the instantaneous direction of motion at each location x and time $t \in [0, 1]$. At inference, integrating the learned ordinary differential equation (ODE) $\dot{x}_t = v_\theta(x_t, t)$ from $t = 0$ to $t = 1$ pushes a fresh noise sample forward into a fresh data sample.

A common choice of probability path is the linear interpolant, $x_t = (1 - t)x_0 + tx_1$ for $x_0 \sim p_0$ and $x_1 \sim p_1$. The model’s target is the *marginal* velocity field at (x, t) : the average direction of motion over all noise-data pairs whose paths pass through that point.

However, supervising this marginal directly is intractable, since the contributing pairs are not known in advance. Flow matching sidesteps this difficulty with a key observation [2]: although the marginal field may be intractable, the *conditional* velocity along the path of any specific noise-data pair is relatively trivial. Indeed, for the linear interpolant, it is simply $\dot{x}_t = x_1 - x_0$. Regressing the model on these conditional velocities, averaged over many sampled pairs, recovers the marginal field in expectation, and gives the training loss as a mean-squared regression:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, x_0, x_1} \|v_\theta(x_t, t) - (x_1 - x_0)\|^2.$$

In practice, every training step samples a time $t \in [0, 1]$ and a noise-data pair (x_0, x_1) , computes the interpolant x_t , and takes a gradient step on this squared error.

Flow matching with this linear interpolant generalises score-based diffusion: the variance-preserving Gaussian-path objective of denoising diffusion models [1] is a special case of the FM family [2, 4].

Generating samples from a trained flow-matching model amounts to integrating the learned velocity field. This is often performed with a standard ODE solver, with Euler steps usually sufficing for stable trajectories. The field also admits stochastic sampling variants — such as those drawing on the connection to score-based stochastic differential equations (SDEs) — which inject noise during the integration steps to explore the probability space and improve sample diversity. The models we work with in this report, Tabasco and Proteina, support both deterministic and stochastic samplers.

Conditional generation augments v_θ with context tokens — such as a class label, a sequence length, or a structural motif — with conditioning supplied at both training and sampling time. In this report, we focus on unconditional generation only.

For a more complete treatment of flow matching, we refer the reader to Lipman et al. [2] for the original presentation, and to the recent Lipman et al. guide [4].

2.2 Representation alignment

Chapter 1 introduced the argument made by Yu et al. [5] about the *representation bottleneck* in diffusion-transformer models. Their analysis applies to flow-matching models as well, given the equivalence between the two training objectives discussed above (§2.1). The flow-matching objective also implicitly asks a model to learn two things at once: a useful representation of the data, and a noise-to-data velocity field that interprets this representation. This twin burden, they argue, contributes to slow convergence and large training budgets.

Their proposed solution, Representation Alignment (REPA), addresses the first half of this burden by adding a regularisation term to the training loss. Figure 2.1 shows the

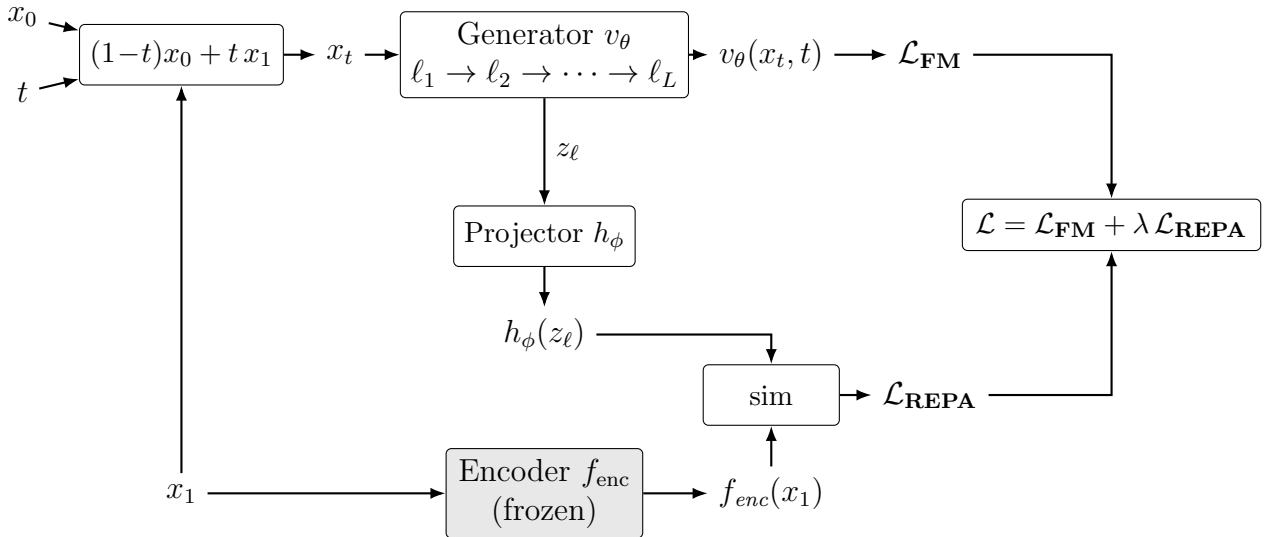


Figure 2.1: The REPA construction. A noised sample $x_t = (1-t)x_0 + tx_1$ is constructed at each training step from a noise sample x_0 , a clean data sample x_1 , and a sampled timestep $t \in [0, 1]$. The generator v_θ processes x_t through its sequence of layers $\ell_1, \dots, \ell_L$ to produce the velocity field that feeds the flow-matching loss $\mathcal{L}_{\text{FM}}$. Separately, the clean sample x_1 is also passed through a frozen pretrained encoder f_{enc} (shaded). The generator’s hidden state z_ℓ at the chosen alignment layer is mapped into the encoder’s feature space by a trainable projector h_ϕ ; the REPA loss $\mathcal{L}_{\text{REPA}}$ is the negated similarity between $h_\phi(z_\ell)$ and $f_{\text{enc}}(x_1)$, averaged across tokens. The total training loss $\mathcal{L}$ combines them with weight λ .

construction. The regularisation pulls z_ℓ , the generator’s intermediate hidden state at a chosen *alignment layer* ℓ , towards the embeddings produced by a frozen pretrained encoder f_{enc} run on the clean data sample x_1 . The hidden state z_ℓ is a sequence of N token vectors. In the context of images, each token represents an image patch; in the molecular setting, each token represents an atom or residue that is part of the larger molecule (§2.3). A small trainable *projector head* h_ϕ — a three-layer multi-layer perceptron (MLP) in the original paper — maps z_ℓ into the encoder’s feature space, so that $h_\phi(z_\ell)$ and $f_{\text{enc}}(x_1)$ can be compared. This alignment is asymmetric: gradients update only the generator parameters θ and the projector ϕ , while the encoder provides a fixed alignment target throughout training. Information thus flows one way, from the pretrained encoder into the generator’s hidden representations.

The alignment loss is the negated similarity, averaged token-wise between projected and target embeddings:

$$\mathcal{L}_{\text{REPA}} = -\mathbb{E}_{t, x_0, x_1} \left[\frac{1}{N} \sum_{n=1}^N \text{sim}(h_\phi(z_\ell)_n, f_{\text{enc}}(x_1)_n) \right],$$

where n indexes the N tokens within a sample. In the image setting, N is constant across the dataset, so this token-wise average is equivalent to averaging similarities once per sample and then across the batch — the original paper draws no distinction between the two. For molecular data, where N varies per sample, the two diverge. We return to this

distinction in the design knobs below.

The final training objective is the flow-matching loss from §2.1 plus this term, scaled by a weight λ :

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda \mathcal{L}_{\text{REPA}}.$$

This construction exposes several design knobs, whose effects we revisit in the ablations of Chapter 7:

- The choice of encoder f_{enc} . Different encoders capture distinct aspects of the data distribution.
- The alignment layer ℓ at which we attach the projector. The optimal choice is not obvious a priori. Yu et al. experimentally determined that mid-network layers work best for image diffusion, but this may not hold in the molecular setting.
- The alignment-loss weight λ , which balances the alignment signal against the generative objective.
- The averaging mode. Averaging tokens within each sample first weights every molecule equally, while pooling tokens across the batch gives larger molecules proportionally more influence.
- The choice of similarity metric sim . Yu et al. compared cosine similarity and mean-squared error for image diffusion and found cosine more effective, a result we corroborate in the molecular setting.

In this report, we adopt REPA in its original form: cosine similarity, additive combination with the flow-matching loss, and a frozen encoder. A growing family of variants relaxes one or more of these choices — for example, encoder-free dispersive regularisers [12] that drop the external encoder entirely, or joint-training extensions that fine-tune the encoder alongside the generator. We defer their discussion to Chapter 3.

Chapter 3

Related work

Chapter 4

Evaluating molecular generation

Chapter 5

Encoder targets and profiling

Chapter 6

REPA on small molecules

Chapter 7

REPA on protein backbones

Chapter 8

Conclusions

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Appendix A

Technical details

A.1 Placeholder section

TODO.